

Raising the night-time efficiency to 65% also raises our overall efficiency, for a ½-day ½-night period, to 67%, and, our satellite example gains as well—jumping to almost 71%. OK, so it only grew 2.3%, but when your system is living on a finite energy source such as solar energy, that small amount could save you the cost of one or two solar cells in a large array.

Even with this change, we still have some exposure to reliability problems. Namely, one of the cells in the multi-cell battery could fail, rendering the battery voltage too low to forward bias the diode against the voltages of the other batteries in the system. Or, one of the solar cells in a string could fail open, leaving the entire string unusable.

Optimally, we could put each solar cell on its own MPP and each cell in the battery on its own charger. However, we would need a boost switcher to raise the battery cell voltage to the input voltage of the load's switching regulators. And, the quiescent and switching currents for all of the new switching regulators would begin to dominate, decreasing our efficiency once again.

The alternative is to balance the needs for efficiency against reliability, by grouping some battery cells and solar cells together, so as to limit losses while also limiting the effect of single-point failures:

- With the solar cells, we have two options. We can diode OR two or more strings of solar cells into a single MPP converter to limit the losses due to the MPP. Or, we can put a single string into multiple MPP converters to eliminate the ORing diode losses. The challenge will be to determine which configuration is more efficient, as well as the optimum number of solar cells per string.

- The batteries require a similar optimisation, to determine how many battery cells in a string will produce a minimum-loss system—given the losses in the charger and boost regulators. This job is simpler, in that we will

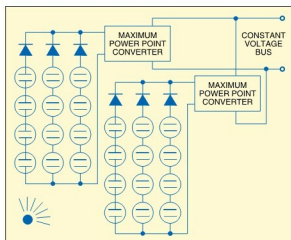


Fig. 5: One possible configuration for the solar/MPP configuration

required to switch the MOSFET's gate. If we drop the current in/out of the MPP, we also reduce the losses in the inductor core, the FR losses in the MOSFETs, and the FR losses due to the ESR of the input and output capacitors, by the square of the current reduction.

Based on this knowledge, we can make some assumptions about increasing efficiency in the system. Shorter strings

Potential Trade-offs

Colour	Semiconductor material
Number of cells in a string	N
Number of strings in a MPP	M
MPP voltage	$(N * V_{cell}) - V_{diode}$
MPP current	$M * I_{cell}$
Number of MPPs	K
MPP voltage losses	If $((N * V_{cell}) - V_{diode}) < 30V$ Then Loss = $K * \text{lowV}$ If $(30V < ((N * V_{cell}) - V_{diode}) < 60V)$ Then Loss = $K * \text{midV}$ If $((N * V_{cell}) - V_{diode}) > 60V$ Then Loss = $K * \text{highV}$
MPP current losses	If $(M * I_{cell}) < 20A$ Then Loss = $K * \text{lowI}$ If $(20A < (M * I_{cell}) < 40A)$ Then Loss = $K * \text{midI}$ If $((M * I_{cell}) - V_{diode}) > 40A$ Then Loss = $K * \text{highI}$
MPP quiescent losses	$K * I_{quies}$
Diode losses	$I_{cell} * M$

only have one string per charger/boost regulator. However, the optimisation is still a trade-off between the number of charge/boost circuits and the losses in each.

Let's start with the efficiency losses in the MPP converter. The biggest losses are in the inductor core, the 'on' resistance of the MOSFETs, the Equivalent Series Resistance (ESR) in the input and output capacitors, and the switching current required to charge and discharge the gate capacitance of the MOSFETs. If we drop the operating voltage, we gain the use of MOSFETs with lower on resistances, and lower gate capacitances. This drops our losses, due to the on resistance of the MOSFET and the switching currents

of cells can increase the efficiency of the MPP converters, by reducing the operating voltage. Longer strings of cells, or fewer shorter-string diodes ORed together, also increase efficiency by reducing the in/out current of the MPP converter. Each MPP converter will have a fixed loss that is dictated by switching losses and the quiescent current of the MPP converter. Balancing these losses, plus the diode loss, against gains in efficiency will be the trade-off. Grouping it all together, we get the above list of equations (see Table) governing the potential trade-offs.

Now, all that is needed is a spreadsheet to automatically recalculate the system loss, based on N, M & K. This exercise is a simple matter of trying dif-